INCH-POUND

MIL-E-49150C(CR)
26 May 1992
SUPERSEDING
MIL-E-49150B(CR)
8 November 1989

MILITARY SPECIFICATION

ENTRANCE HOUSING ASSEMBLY VIEWER, DRIVER'S, NIGHT VISION, AN/VVS-2(V)

This specification is approved for use by US Army Communications-Electronics Command, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the Entrance Window and Mirror Housing Assembly which is part of Viewer, Driver's Night Vision, AN/VVS-2(V).

1.2 Classification. Entrance assemblies covered by this specification shall be the following types as specified;

Type I Entrance Window and Mirror Housing Assembly for AN/VVS-2(V)1A. (M60 application.)
Type II Prism Objective Housing Assembly for AN/VVS-2(V)2A (XM-1 application) and AN/VVS-2(V)3 (FVS application).

2. APPLICABLE DOCUMENTS

2.1 Government documents.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving the document should be addressed to: HQ, USA Communications-Electronics Command, ATTN: AMSEL-ED-TM Fort Monmouth, NJ 07703-5023 by using the Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 5855

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2.1.1 Specifications and standards. The following specifications and standards form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2.e).

SPECIFICATIONS

MILITARY

MIL-P-11268	-	Parts, Materials, and Processes Used in Electronic Equipment
MIL-M-13231	-	Marking of Electronic Items
MIL-O-13830	-	Optical Components for Fire Control Instruments; General Specification Governing the Manufacture, Assembly, and Inspection of
MIL-F-14072	-	Finishes for Ground Electronic Equipment

STANDARDS

MILITARY

MIL-STD-150	-	Photographic Lenses
MIL-STD-810	-	Environmental Test Methods

(Unless otherwise indicated, copies of federal and military specifications, standards and handbooks are available from the Standardization Document Order Desk, 700 Robbins Avenue, Building #4, Section D, Philadelphia, PA 19111-5094.)

2.1.2 Other Government documents, drawings and publications. The following other Government documents, drawings and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation (see 6.2.e).

DRAWINGS

USA COMMUNICATIONS-ELECTRONICS COMMAND

SM-D-771333	-	Entrance Window and Mirror Housing Assembly
SM-D-771434	-	Prism Objective Housing Assembly

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

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2.2 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DOD adopted are those listed in the issue of the DODISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS are the issues of documents cited in the solicitation (see 6.2.e).

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ATSM)

ASTM B26	Aluminum Alloy Sand Castings
ASTM B108	Aluminum Alloy Permanent Mold Castings

(Application for copies should be addressed to the American Society for Testing Materials, 1916 Race Street, Philadelphia, PA 19103-1187.)

2.3. Order of precedence. In the event of a conflict between the text of this specification and the references cited herein (except for associated detail specification, specification sheets or MS standards), the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Description. The entrance window and mirror housing assembly is herein referenced to as the entrance assembly. The entrance assembly is the portion of the driver's viewer outside the vehicle. It directs the line of sight to the objective lens.

3.2 Construction. The entrance assembly shall be as specified herein and in accordance with the following data lists as applicable (see 6.2.d):

Type I	SM-D-771333 Entrance Window and Mirror Housing Assembly
Type II	SM-D-771434 Prism Objective Housing Assembly

3.3 Qualification. Unless otherwise specified by the contract or purchase order, Housing Assemblies furnished under this specification shall be products approved as qualified products (see 3.12.3 and 6.3).

3.3.1 Initial production testing (IPT). When specified in the contract (see 6.2.c) the contractor shall furnish unit(s) for IPT inspection (see 4.3).

3.4 Parts and materials. Parts and materials shall be as specified herein and as shown on applicable drawings. Materials not specified shall be selected by the contractor and shall be subject to all provisions of this specification and shall conform to MIL-P-11268.

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3.5 Performance characteristics.

3.5.1 Angular deflection. An optical ray perpendicular to the mounting flange shall be deflected through an angle of 108, ± 2 degrees for Type I and 90, ± 2 degrees for Type II.

3.5.2 Resolution. The resolution shall be 7 line-pairs/milliradian or better, using a telescope of 8 power (X) or greater with a 30 mm aperture.

3.5.3 Transmission. The transmission shall be 80 percent minimum over the entire aperture in the spectral region defined by a 2854 K, ± 200 K source and an S-20 extended-red (ER) detector (see Figure 1).

3.6 Environmental.

3.6.1 High temperature. The assembly shall withstand storage temperature of +68°C without damage or degradation of performance (see 3.12.1 and 3.12.5).

3.6.2 Low temperature. The assembly shall withstand storage temperature of -51°C without damage or degradation of performance (see 3.12.1 and 3.12.5).

3.6.3 Temperature shock. The entrance assembly shall withstand cycles of sudden changes in temperature between -51°C and +68°C without damage or degradation of performance (see 3.12.1 and 3.12.5).

3.6.4 Humidity. The entrance assembly shall withstand +68°C and 94, ±4 percent humidity without damage or degradation of performance (see 3.12.1 and 3.12.5).

3.6.5 Sealing. The entrance assembly shall be filled with a mixture of dry nitrogen with 1 mole-percent helium and shall be sealed. The helium leak rate of the sealed assembly shall be not greater than 9.9×10^{-4} cm^3/sec. The entrance assembly shall withstand sealing test without damage or degradation of performance (see 3.12.1 and 3.12.5).

3.6.6 Vibration. The entrance assembly shall withstand vibration equivalent to that encountered in tracked vehicles during 5000 miles of operation without damage or degradation of performance (see 3.12.1 and 3.12.5).

3.6.7 Shock.

3.6.7.1 Basic shock. The entrance assembly shall withstand a basic shock of 40g's (see 3.12.6) with a duration of 18 milliseconds without damage or degradation of performance (see 3.12.1 and 3.12.5).

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3.6.7.2 Weapon shock. The entrance assembly shall withstand a simulated weapon firing shock of 100 g's with a duration of 1.5 milliseconds one half sine pulse, without damage or degradation of performance (see 3.12.1 and 3.12.5).

3.7 Treatment and painting. Unless otherwise specified, treatments and painting shall conform to MIL-F-14072.

3.8 Marking. Unless otherwise specified, the entrance assembly shall be marked and identified per MIL-M-13231.

3.9 Castings. Unless otherwise specified, castings shall conform to ASTM B26 or ASTM B108.

3.10 Cleanliness. Entrance assembly shall meet the cleanliness requirement for optical systems as specified in MIL-O-13830.

3.11 Workmanship. Standards of workmanship shall be such that components will meet all requirements of this specification and any referenced specifications or drawings where inspected in accordance with Section 4. Parts and assemblies shall comply with the requirements of MIL-O-13830.

3.12 Technical interpretations. The following technical interpretations are, when referenced in sections 3, 4, or 5, mandatory for this specification.

3.12.1 Damage. Damage is defined as any breakage, loosening, shift, dents, failure of any finish, hardware, connection, or component, or condensation, fogging or moisture on any internal surface.

3.12.2 Ambient temperature. Ambient temperature is defined as +23°C, ± 10°C.

3.12.3 Qualified product. A product that has successfully met all the requirements of an IPT or First Article witnessed by the CECOM, Concurrent Engineering Directorate or its designated representative. Should the supplier make substantial changes in the process, personnel, location, or equipment used to produce a qualified product, or have not delivered the qualified product to the Federal Government within 12 months of contract award, an IPT shall be performed to the extent necessary to verify continued compliance to specification requirements for all characteristics of the product affected by the changes.

3.12.4 Environmental tests. For the purpose of this specification environmental test are those test specified in 4.6.

3.12.5 Degradation of performance. Any deterioration in quality, or failure to operate in accordance with the requirements of this specification.

3.12.6 "g". "g" is a unitless measure of the quantity of units of force associated with the earths gravity that a object is subjected to.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items shall meet requirements of section 3 and 5. The inspection set forth in this specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspections, as part of manufacturing operations, is an acceptable practice to ascertain conformance to requirements, however, this does not authorize submission to known defective material, either indicated or actual, nor does it commit the Government to accept defective material.

4.2 Classification of inspection. Inspection shall be classified as follows:

a. Initial production testing (4.3).
b. Quality conformance inspection (4.4).
c. Inspection for packaging (4.7).

4.3 Initial production testing.

4.3.1 Inspection. Each IPT entrance assembly shall be inspected in accordance with Table I, either in process or when assembled, as required. Presence of one or more defects shall be cause for rejection of that assembly and may be cause for rejection of IPT. Unless otherwise specified (see 6.2.c), ten entrance assemblies manufactured with production tooling shall be selected from the first production lot and provided for IPT.

TABLE I. Inspection.

Inspection	Requirement paragraph
Any part or materials not as specified	3.4
Treatment and painting not as specified	3.7
Markings not as specified or illegible	3.8
Castings	3.9
Cleanliness	3.10
Workmanship	3.11

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4.3.2 IPT tests. Following successful completion of the inspections specified in 4.3.1 each IPT entrance assembly shall be subjected to all the tests in Table II, which may be performed in any order. Failure of any test shall be cause for rejection of the IPT.

4.3.2.1 Disposition of IPT samples. Disposition of IPT samples shall be as specified in the contract or purchase order.

TABLE II. Initial production testing.

Inspection	Requirement paragraph	Test paragraph
Angular deflection	3.5.1	4.5.1
Resolution	3.5.2	4.5.2
Transmission	3.5.3	4.5.3
High temperature	3.6.1	4.6.1
Low temperature	3.6.2	4.6.2
Temperature shock	3.6.3	4.6.3
Humidity	3.6.4	4.6.4
Sealing	3.6.5	4.6.5
Vibration	3.6.6	4.6.6
Shock	3.6.7	4.6.7

4.4 Quality conformance inspection.

4.4.1 Examination. Each entrance assembly shall be examined as specified in Table I. Presence of one or more defects shall be cause for rejection.

4.4.2 Test.

4.4.2.1 Group A inspection. Entrance assemblies that have passed the examination specified in 4.4.1 shall be tested in accordance with Table III. Failure of any test shall be cause for rejection of that unit. Entrance assemblies which pass Group A inspection shall be formed into lots by the contractor.

TABLE III. Group A inspection.

Inspection	Requirement paragraph	Test paragraph
Angular deflection	3.5.1	4.5.1
Resolution	3.5.2	4.5.2

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4.4.2.2 Group B inspection. Group B inspection shall be conducted on entrance assemblies from lots which have passed the examination and the Group A test. Sampling shall be in accordance with contract or purchase order (see 6.2.f). Tests shall be conducted in accordance with Table IV.

TABLE IV. Group B inspection.

Inspection	Requirement paragraph	Test paragraph
Transmission	3.5.3	4.5.3

4.4.2.3 Group C inspection. This inspection shall consist of the tests specified in Table V and shall be performed on entrance assemblies from lots which have passed Group A inspection.

4.4.2.4 Sampling of Group C inspection. This inspection shall be as specified in the contract or purchase order (see 6.2.f).

TABLE V. Group C inspection.

Inspection	Requirement paragraph	Test paragraph
Subgroup 1		
Vibration	3.6.6	4.6.6
High temperature	3.6.1	4.6.1
Low temperature	3.6.2	4.6.2
Subgroup 2		
Temperature shock	3.6.3	4.6.3
Shock	3.6.7	4.6.7
Sealing	3.6.5	4.6.5
Subgroup 3		
Humidity	3.6.4	4.6.4

4.4.2.5 Group C failures. Actions required relative to Group C failures shall be as specified in the contract or order (see 6.2.b).

4.4.2.6 Disposition of Group C samples. Unless otherwise specified, samples which have been subjected to Group C tests shall be refurbished, pass the requirements of Group A tests and shipped against the requirements of the contract.

4.5 Test procedure. Unless otherwise specified all tests shall be performed at ambient temperature (see 3.12.2).

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4.5.1 Angular deflection. The assembly is placed in the optical path of a collimator with the mounting flange perpendicular to the beam (see Figure 2). The angular deflection of this beam through the window is measured. Failure to meet requirements of 3.5.1 shall constitute failure of this test.

4.5.2 Resolution. The entrance assembly, when inserted in the path of an 8X telescope with a 30 mm aperture, see Figure 2, shall be capable of resolving at least 7 lp/milliradian when the assembly is rotated ± 15 degrees about a center 35 mm from the mounting flange. The telescope may be focused a maximum of 0.03 diopter from infinity for a best focus. Failure to meet the requirements of 3.5.2 shall constitute failure of this test.

4.5.3 Transmission. The transmission of the assembly shall be measured in accordance with MIL-STD-150 except that the source shall be 2854K, ± 200K and the detector shall be S-20 extended red (see Figure 1). Failure to meet the requirements of 3.5.3 shall constitute failure of the test.

4.6 Environmental. The entrance assembly shall be retested to meet the requirements of 3.5.1, 3.5.2, and 3.5.3 after the completion of each environmental test (see 3.12.4). Failure to meet these requirements shall constitute failure as defined in degradation of performance (see 3.12.5). Temperature chambers shall maintain specified temperatures within ± 2.0°C.

4.6.1 High temperature. The entrance assembly shall be subjected to the test Method 501.1, Procedure I of MIL-STD-810, except that the temperature of Steps 2 and 3 (highest storage temperature) shall be +68°C and Step 4 and 5 shall be deleted. Step 6 stabilization period shall be a minimum of two hours. Failure to meet the requirements of 3.5.1, 3.5.2, 3.5.3, and 3.6.1 shall constitute failure of this test.

4.6.2 Low temperature. The entrance assembly shall be subjected to the test Method 502.1, Procedure I of MIL-STD-810. The storage temperature (Step 2) shall be -51°C. Steps 2 and 6 stabilization period shall be a minimum of two hours, and Steps 4 and 5 shall be deleted. Failure to meet the requirements of 3.5.1, 3.5.2, 3.5.3, and 3.6.2 shall constitute failure of this test.

4.6.3 Temperature shock. The entrance assembly shall be subjected to the test Method 503.1, Procedure I of MIL-STD-810, except the high temperature chamber shall be maintained at +68°C, the low temperature chamber shall be maintained at -51°C, and Steps 1, 3, and 5 stabilization period shall be a minimum of two hours. Step 1 and 4 shall be at 68°C, step 2 shall be at -51°C. Immediately after each cold cycle the assembly shall be examined for internal condensation. Evidence of moisture or condensation on any internal surface or failure to meet the requirements of 3.5.1, 3.5.2, 3.5.3, and 3.6.3 shall constitute failure of this test.

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4.6.4 Humidity. The entrance assembly shall be subjected to the test Method 507.1,Procedure II of MIL-STD-810. Failure to meet the requirements of 3.5.1, 3.5.2, 3.5.3, and 3.6.4 shall constitute failure of this test.

4.6.5 Sealing. The entrance assembly shall be subjected to the test of Method 512.1, Procedure IV of MIL-STD-810. The entrance assembly shall then be tested for helium leakage with a Varian Model 925-40 Porta-Test or equivalent. Failure to meet the requirements of 3.5.1, 3.5.2, 3.5.3, and 3.6.5 shall constitute failure of this test.

4.6.6 Vibration. The entrance assembly shall be subjected to MIL-STD-810, Method 514.2, Procedure VIII, Table 514.2-6, Tracked Vehicles, limited to one sweep in the vertical axis only. Failure to meet the requirements of 3.5.1, 3.5.2, 3.5.3, and 3.6.6 shall constitute failure of this test.

4.6.7 Shock. The entrance assembly shall be subjected to the test of MIL-STD-810, Method 516.2.

a. 40 g's at 18 milliseconds duration, Figure 516.2-1, shock procedure I.

b. Weapon firings, 100 g's using procedure IV except at 1.5 milliseconds (ms) duration, Figure 516.2-2.

Failure to meet the requirements of 3.5.1, 3.5.2, 3.5.3, and 3.6.7 shall constitute failure of this test.

4.7 Inspection for packaging. Packing and packaging shall be inspected in accordance with requirements of the contract or purchase order (see 6.2.h).

5. PACKAGING

5.1 Packaging requirements. The packaging requirements for the desired level(s) of protection shall be as specified by the acquisition activity (see 6.2.h).

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The entrance assembly is used with the AN/VVS-2(V) for driver's viewing during period of darkness.

6.2 Acquisition requirements. Acquisition documents must specify the following:

a. Title, number and date of this specification.
b. Necessary actions by contractor in the event of a Group C failure (see 4.4.2.5).

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c. Quantity and schedule for Initial Production testing for all line items and disposition of IPT samples.
d. Type of entrance assembly (see 3.2).
e. Issue of DODISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.1.1, 2.1.2 and 2.2).
f. Sampling plans for Group B and Group C testing. As guidance, unless otherwise specified, sampling shall be conducted per the requirements of MIL-STD-105.
g. MIL-STD-810C is used for environmental tests of section 4.
h. Packaging requirements.

6.3 Qualification. With respect to products requiring qualification, unless otherwise specified by the contract or purchase order, if a product is not qualified at time of award, the contract shall require qualification prior to first delivery. The activity responsible for approving qualified products is the CECOM, Concurrent Engineering Directorate, ATTN: AMSEL-ED-IN, Fort Monmouth, NJ 07703. Information pertaining to qualification of products may be obtained from the above activity.

6.4 Definitions. See 3.12.

6.5 Subject term keyword listing.
Entrance Assembly
Viewer Driver's

6.6 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

Custodian:
Army - CR

Preparing Activity:
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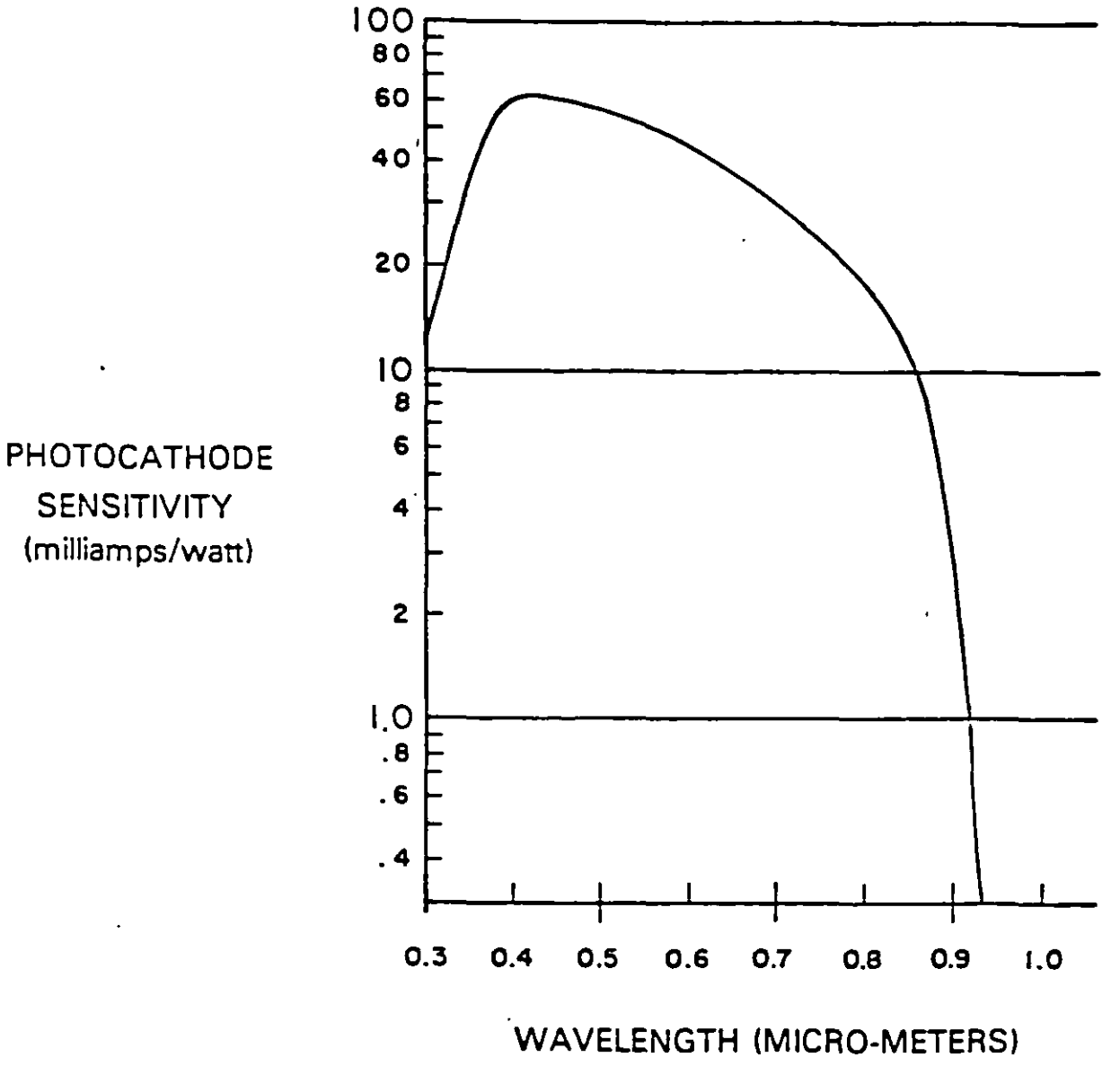


FIGURE 1. RESPONSE CURVE, S-20 EXTENDED RED PHOTOCATHODE

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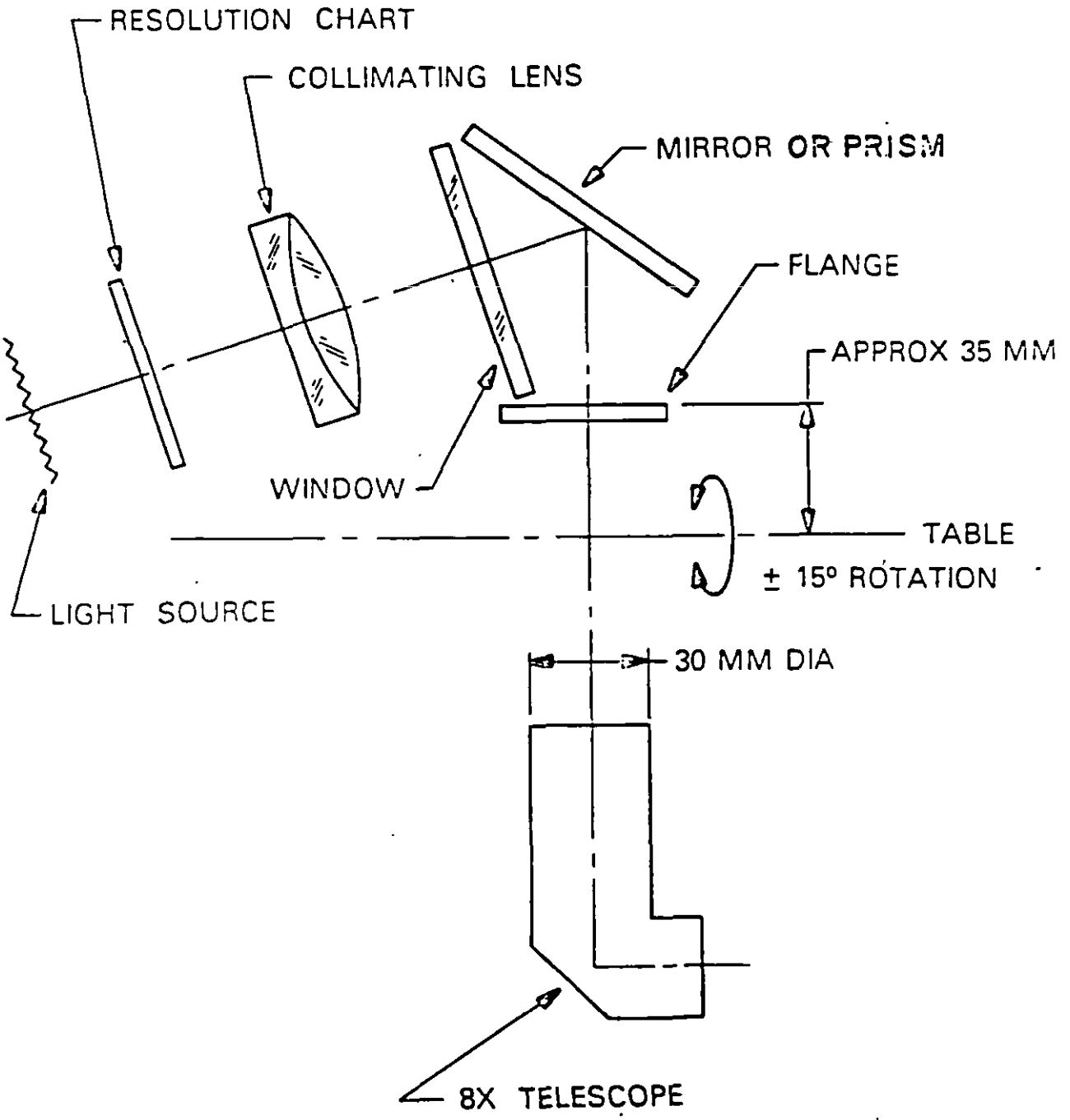


FIGURE 2. ENTRANCE WINDOW HOUSING TESTS DIAGRAM

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.
2. The submitter of this form must complete blocks 4, 5, 6, and 7.
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NOTE: This form may not be used to request copies of documents, nor to request waivers, or clarification of requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER	2. DOCUMENT DATE (YYMMDD)
	MIL-E-49150C(CR)	26 May 1992

3. DOCUMENT TITLE
ENTRANCE HOUSING ASSEMBLY VIEWER, DRIVER'S NIGHT VISION, AN/VVS-2(V)

4. NATURE OF CHANGE *(Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)*

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (Last, First, Middle Initial)	b. ORGANIZATION	
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON (If applicable)	7. DATE SUBMITTED (YYMMDD)

8. PREPARING ACTIVITY

a. NAME	b. TELEPHONE (Include Area Code) (1) Commercial	(2) AUTOVON
CECOM	(908) 532-5851	992-5851
c. ADDRESS (Include Zip Code) AMSEL-ED-TM Fort Monmouth, NJ 07703-5023	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 . AUTOVON 289-2340	

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